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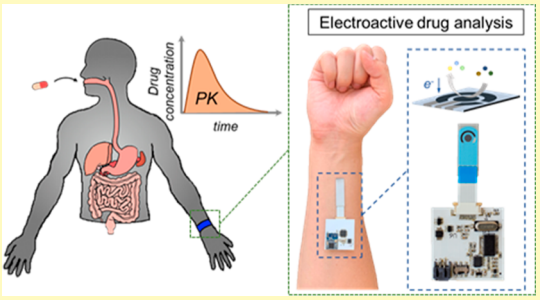
# Design Framework and Sensing System for Noninvasive Wearable Electroactive Drug Monitoring

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Supporting Information

**ABSTRACT:** Wearable drug monitoring targeting epidermally retrievable biofluids (e.g., sweat) can enable a variety of applications, including drug compliance/abuse monitoring and personalized therapeutic drug dosing. In that regard, voltammetry-based approaches are suitable because they uniquely leverage the electroactive nature of target drug molecules for quantification, eliminating the reliance on the availability of recognition elements. However, to adapt such approaches for the envisioned application, three main challenges must be addressed: (1) constructing a sensitive voltammetric sensing interface with high signal-to-background ratio, (2) decoupling the confounding effect of endogenous electroactive species (naturally present in complex biofluid matrices) and baseline variation, and (3) realizing wireless voltammetric excitation and signal acquisition/transmission. To this end, first, a framework for the quantification of electroactive drugs is presented, which centers on the evaluation and determination of suitable sensing electrodes and characterization of the interference from a panel of physiologically relevant electroactive species. This framework was utilized to establish the design space and operational settings for the development of a coupled sensing system and analytical framework to render sample-to-answer drug readouts in complex biofluid matrices. The presented design framework and sensing system can serve as a basis for future wearable sensor development efforts aiming to monitor electroactive species such as pharmaceutical molecules.




**KEYWORDS:** *wearable drug monitoring, personalized dosing, differential pulse voltammetry, electroactive species, boron-doped diamond electrode, analytical framework*

Development of microfluidic devices, biochemical sensing techniques, and miniaturized electronics has enabled the (semi)continuous monitoring of clinically informative endogenous and exogenous molecules in epidermally retrieved biofluids (e.g., sweat).[1−4] Previous work focused on devising electrochemical sensing interfaces and systems to target endogenous biomarkers (e.g., metabolites and electrolytes),[5−8] but exogenous molecules, such as drugs, remain underexplored. Given the low-molecular-weight (<a few kDa) and high-lipophilicity properties of most drug molecules, these analytes are speculated to partition into noninvasively retrievable biofluids, such as sweat, in correlation with blood.[9−11] Therefore, in principle, sweat analysis can be used to provide proxy measures of target drug concentration level in blood noninvasively, potentially opening up new directions in therapy management, including drug compliance/abuse monitoring, drug−drug interaction study, and personalized dosing (Figure 1a).

A number of challenges with regards to the development of wearable sensing interfaces for reliable drug quantification must be considered and addressed: (1) because of the exogenous nature of the drugs, in general, biological recognition elements (e.g., enzyme, antibody) are not naturally available to construct the sensing interface and (2) drug molecules typically appear at low concentration levels in biofluids, rendering their quantification in complex biofluid matrix (containing highly abundant non-target/interfering species) difficult. Given the context at hand, voltammetry-based approaches can be adopted to construct drug-sensing interfaces, because they uniquely leverage the electroactive nature of target drug molecules for quantification, eliminating the reliance on the availability of recognition elements. Specifically, pulse voltammetry (including differential pulse voltammetry, DPV, and square wave voltammetry) is widely utilized for the quantification of electroactive species, because of their ability to suppress non-faradaic background current.[12] The quantification principle of these methods is based on

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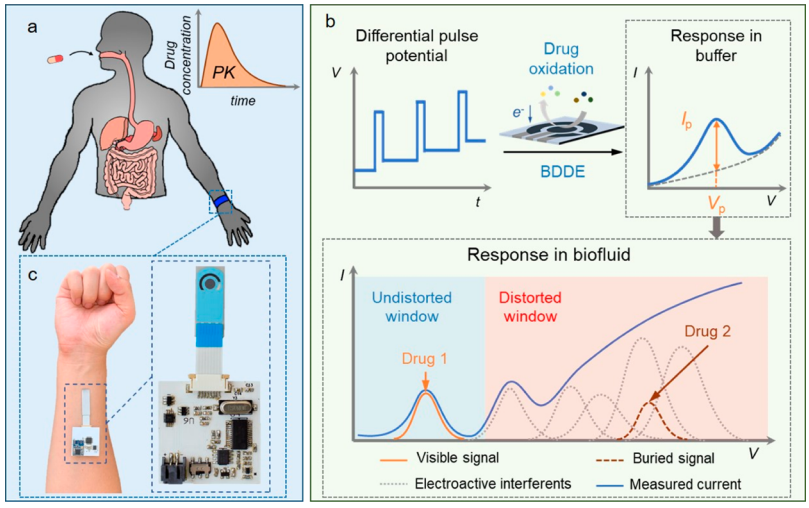



**Figure 1.** Conceptual illustration of the design framework for electroactive drug monitoring. (a) Wearable drug sensing system, which can be applied for pharmacokinetic (PK) monitoring. (b) Voltammetric drug analysis. The upper part shows the schematic for DPV-based drug analysis using BDDE, which, in principle, can capture the oxidation of the electroactive drug molecules (manifested as a current peak at the drug's oxidation fingerprint potential in voltammogram). The lower part illustrates the distortion in the voltammogram (in a complex biofluid) because of the interference of endogenous electroactive species. Analytes electro-oxidized in the undistorted potential window (e.g., drug 1) can be selectively and sensitively detected, whereas the fingerprint peak of the analytes oxidized in the distorted potential window (e.g., drug 2) may be buried in the background. (c) Developed wearable voltammetric drug-sensing system (a BDDE-based sensor interfacing a DPV readout circuit board) mounted on a subject's forearm.

sweeping voltage (applied across the working and reference electrodes) and recording the characteristic current peak at the analyte's fingerprint redox voltage—in which case the measured peak current is proportional to the target analyte's concentration level (assuming no contribution from other electroactive species).

To adapt voltammetry-based analysis for wearable drug monitoring applications, three main challenges need to be addressed at the sensor and system levels. First, at the sensor level, a sensitive and inherently stable sensing interface with high signal-to-background ratio is required to produce a measurable response when targeting drug molecules within their physiological relevant concentration range (low/sub-micromolar levels[13]) and in native biofluid conditions (e.g., pH). Second, all the electroactive species within the biofluid can contribute to the voltammetric current, where their contributions are effectively superimposed, potentially leading to the burial of the target drug's characteristic peak in the measured voltammetric current. Accordingly, the interfering influence of endogenous electroactive species (naturally present in a complex biofluid matrix) must be carefully examined and decoupled from the measured current. Third, at the system level, to achieve sample-to-answer measurement, wireless voltammetric excitation and signal acquisition/transmission capabilities need to be realized within a small footprint, which necessitates the miniaturization of bulky lab instruments.

In this work, first, a framework is presented, which centers on the evaluation and determination of suitable sensing electrodes for voltammetric analysis and characterization of endogenous interferents. Three model drugs, dipyridamole (DP), acetaminophen (APAP), and caffeine (CAFF) were targeted, which can be electrochemically oxidized at low (<0.5 V), medium (0.5−1 V), and high (>1 V) potentials, respectively.[14−16] These drugs were selected, in particular, because of their significance in disease treatment and the demonstrated utility of monitoring them for improving patient outcomes. For example, DP is used for the cardiovascular diseases treatment because of its vasodilating and antiplatelet properties, and monitoring of antiplatelet therapies has been demonstrated to be critical for stroke patients.[17,18] APAP is a widely used pain reliever and fever reducer with remarkable variation in metabolism, and therapeutic monitoring can help to control the patients' exposure to its potentially toxic metabolite.[19] CAFF is used in the treatment of airway obstruction and prematurity apnea, and therapeutic monitoring has shown to be useful in individualizing dosage to reduce drug toxicity.[20,21]

Based on the examination of the voltammetric performance of carbon-based electrode candidates (which, in principle, can characterize the oxidation of drug molecules), the boron-doped diamond electrode (BDDE) was determined to be a suitable sensing electrode owing to its wide electrochemical potential window and high operational stability. With a deliberate surface treatment of the BDDE, nano- to sub/low-micromolar level quantification capability was realized for all the target drugs. Furthermore, the interference of a panel of physiologically relevant endogenous electroactive species [including uric acid (UA) and amino acids] was characterized in relation to the quantification of the target drugs.

The insight from the studies pursued within the presented framework (Figure 1b) informed the design space and operational settings for the development of a fully integrated voltammetric sensing system (consisting of a BDDE-based sensor for sensitive voltammetric analysis and a wireless DPV readout circuitry for signal acquisition/transmission, Figure 1c) augmented with an analytical framework (for voltammetric signal extraction). The utility of this coupled sensing system and analytical framework to render sample-to-answer drug readouts was validated by quantifying the drug content of spiked sweat samples (iontophoretically induced and spiked with known DP drug levels). The presented design framework can serve as a basis for the future wearable sensor development


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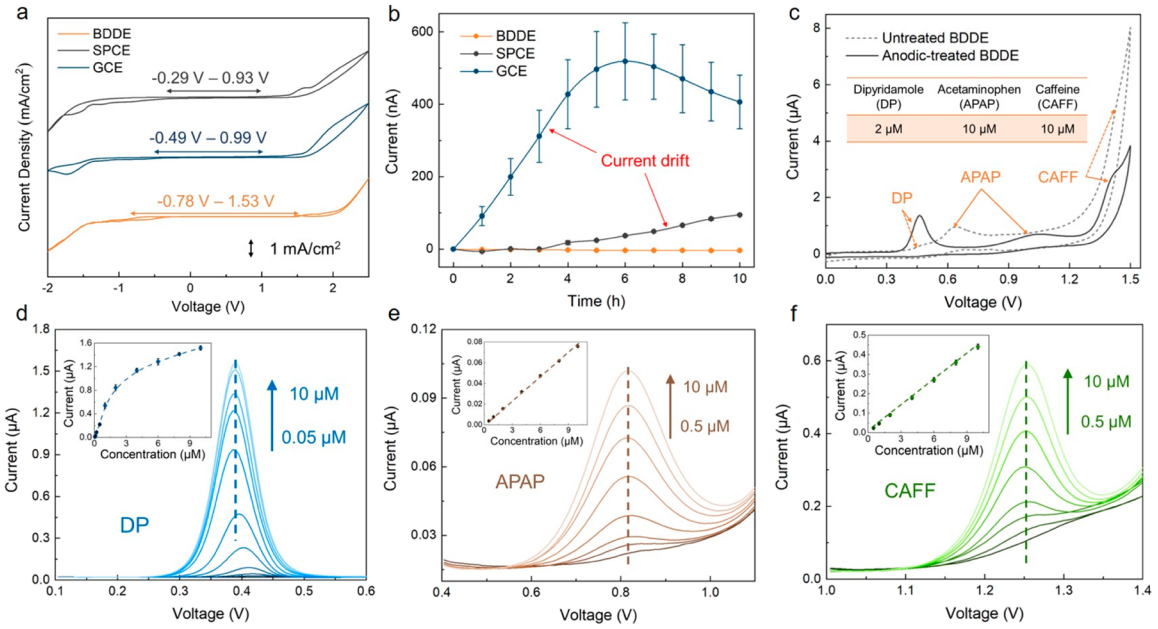



**Figure 2.** (a) Cyclic voltammograms of GCE, SPCE, and BDDE (performed in 0.01 M acetate buffer, pH = 5, scan rate: 25 mV/s) with the annotated potential window. (b) Baseline current of GCE, SPCE, and BDDE under high potential operation (1.2 V vs Ag/AgCl, $n$ = 3, error bars indicate standard error). (c) Cyclic voltammograms of untreated and anodic-treated BDDE in the presence of 2 $\mu$M DP, 10 $\mu$M APAP, and 10 $\mu$M CAFF (performed in 0.01 M acetate buffer, pH = 5, scan rate: 0.1 V/s). (d–f) Differential pulse voltammograms of anodic-treated BDDE with (d) 0, 0.05, 0.1, 0.2, 0.5, 1, 2, 4, 6, 8, 10 $\mu$M DP; (e) 0, 0.5, 1, 2, 4, 6, 8, 10 $\mu$M APAP; and (f) 0, 0.5, 1, 2, 4, 6, 8, 10 $\mu$M CAFF (performed in 0.01 M acetate buffer, pH = 5). Insets show the corresponding calibration curve ($n$ = 3, error bars indicate standard error).

efforts aiming to monitor electroactive species such as pharmaceutical molecules.

## EXPERIMENTAL SECTION

**Materials and Reagents.** All reagents were purchased from Sigma-Aldrich (MO, USA) unless stated. 0.01 M acetate buffer (pH = 5, similar to sweat[22]) was diluted from sodium acetate buffer solution (3M, MN, USA, pH = 5.2 ± 0.1). Artificial eccrine perspiration (stabilized, pH = 5) was purchased from Pickering Laboratories, Inc. (CA, USA).

**Electrode Preparation and Electrochemical Measurements.** The electrochemical measurements were performed with a CHI660E electrochemical workstation (CH Instruments, Inc., TX, USA) BDDE sensor (reference electrode: silver, Ag, counter electrode: carbon, C), purchased from Metrohm USA (FL, USA). Anodic treatment was performed by applying a potential of +2 V (vs silver/silver chloride, Ag/AgCl) for 5 min in 0.5 M sulfuric acid ($H_2SO_4$). Electrochemical cleaning was performed on BDDE each time before use by 10 cycles cyclic voltammetry (CV) scanning (−0.5 to 2.8 V vs Ag/AgCl, scan rate: 0.5 V/s). The glassy carbon electrode (GCE) was purchased from Bioanalytical Systems, Inc. (IN, USA). Each time before use, the GCE was pretreated by polishing with diamond suspension (1 $\mu$m), followed by ultrasonication in ethanol (5 min) and deionized (DI) water (5 min). The screen-printed carbon electrode (SPCE) was purchased from CH Instruments Inc. Characterization of carbon electrodes and sensor calibration were performed, using a CHI660E potentiostat, in an electrochemical cell (0.01 M acetate buffer, pH = 5, reference electrode: Ag/AgCl, counter electrode: platinum, Pt). DPV measurements were performed by drop-casting with increment: 5 mV, amplitude: 50 mV, pulse width: 0.1 s, sampling width: 16.7 ms, and pulse period: 0.5 s.

**Sweat Collection and Measurement.** Sweat collection was performed in compliance with the protocols that have been approved by the Institutional Review Board (IRB) at the University of California, Los Angeles (IRB#17-000170). All subjects gave written informed consent before participation in the study. Standard iontophoresis protocol was followed for sweat stimulation and collection using a Macroduct Sweat Collection System (ELITech Group, France). Accordingly, the volar surface of the human subject's forearm was cleaned with isopropyl alcohol and DI water. Next, 0.5%-pilocarpine-loaded hydrogels were placed onto the designated disk electrodes and mounted onto the cleaned skin surface. By applying 1.5 mA over 5 min (with the aid of a Webster sweat inducer), agonist molecules were delivered underneath the skin to locally stimulate sweat glands. The subsequently secreted sweat was collected with the Macroduct sweat collector. Sweat samples were stored at −20 °C before use. Before analysis, the sweat sample was aliquoted and spiked with different volumes of DP stock solution (100 $\mu$M, in 0.01 M acetate buffer, pH = 5) to the respective expected concentration. Electrochemical cleaning of BDDE was performed after each sweat measurement.

**Wireless DPV Readout System.** The wireless DPV readout circuit board was constructed on a miniaturized flexible printed circuit board. At its core, a microcontroller unit (MCU, ATmega8, Microchip Technology, AR, USA) was utilized to program the required excitation potential waveform and retrieve the readout signal. Serial peripheral interface protocol was followed for communication. The excitation potential waveform (with the same parameters as the potentiostat measurements) was applied across the working and reference electrodes through a 16-bit digital-to-analog converter (DAC, DAC8552, Texas Instrument). The current response from the working electrode was extracted by a transfer impedance amplifier stage (TIA, OPA625, Texas Instrument, TX, USA), which transferred the current signal to a voltage readout. A 24-bit analog to digital converter (ADC, ADS1256, Texas Instruments) was utilized for voltage readout, which was further transmitted to an electrically erasable programmable read-only memory for temporary storage. The circuit board was communicating wirelessly and bilaterally with the user interface through an onboard bluetooth module (HM-11, Seeed Technology Co., Ltd., China). The electrical connections between the BDDE-based sensor and the voltammetric readout board can be established with the aid of a flat flexible cable (Molex, IL, USA) and a double-sided adhesive anisotropic conductive film (9703, 3M). The compatibility of the BDDE for the integration with sweat sampling/routing microfluidic interfaces is shown in Figure S1.



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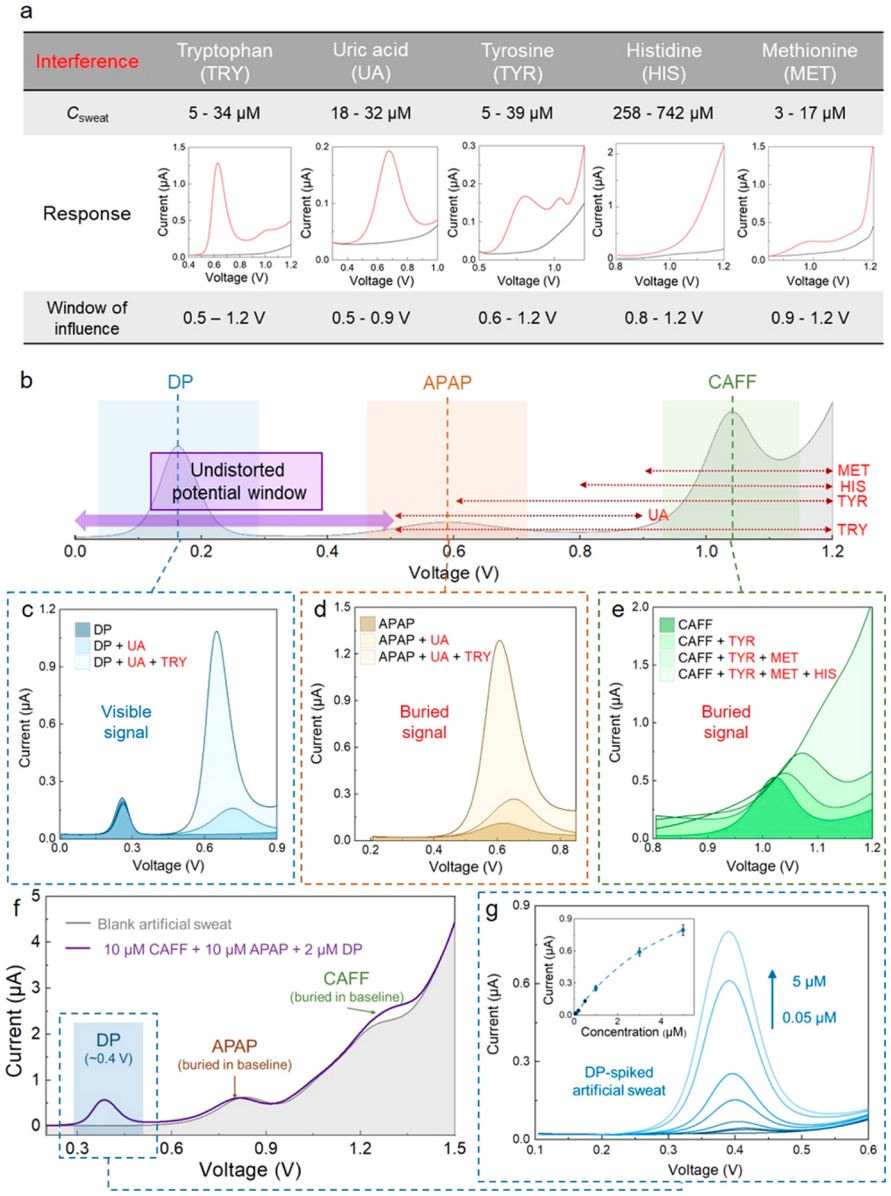



**Figure 3.** (a) Selected panel of endogenous electroactive interferents in sweat. 1st row: the interferents' concentration range in sweat; 2nd row: corresponding differential pulse voltammograms of 20 $\mu$M TRY, 25 $\mu$M UA, 20 $\mu$M TYR, 10 $\mu$M MET, and 500 $\mu$M HIS (red) compared to the respective electrochemical background (black, no interferent spiked, all measurements were performed in 0.01 M acetate buffer, pH = 5); 3rd row: corresponding potential window of influence for each interferent. (b) Schematic differential pulse voltammogram where all the selected model drugs are present. The interferents' potential windows of influence and the undistorted potential window are annotated. (c–e) Differential pulse voltammograms of (c) 2 $\mu$M DP, (d) 10 $\mu$M APAP, and (e) 10 $\mu$M CAFF with subsequent addition of electroactive interferents oxidized at the proximal potential (20 $\mu$M TRY, 25 $\mu$M UA, 20 $\mu$M TYR, 10 $\mu$M MET, and 500 $\mu$M HIS). (f) Differential pulse voltammograms of blank artificial sweat and drug-spiked (2 $\mu$M DP, 10 $\mu$M APAP, and 10 $\mu$M CAFF) artificial sweat (pH = 5). (g) Differential pulse voltammograms of 0, 0.05, 0.1, 0.2, 0.5, 1, 3, 5 $\mu$M DP in artificial sweat (pH = 5). The inset shows the corresponding calibration curve ($n$ = 3, error bars indicate standard error).

**Analytical Framework for Peak Height Information Extraction.** To extract the peak height information from the voltammogram, the residual current outside the peak region was fitted using the combination of a third-order polynomial and exponential equation with proper initial values for all the unknown variables (MATLAB)

$$I_{\text{baseline}} = a_1V^3 + a_2V + a_3 + a_4 \times \exp(a_5V) \quad (1)$$

The peak current was extracted by subtracting the derived background from the current measurements.

## RESULTS AND DISCUSSION

**Determination of Suitable Sensing Electrode for Electroactive Drug Analysis.** To adapt voltammetry for drug-monitoring applications, first, a sensitive and inherently stable sensing electrode needs to be selected. To this end, carbon electrodes are particularly suitable because of their wide electrochemical potential window, which allows for the characterization of drug molecules with high oxidation potential (>1 V).[23] To determine a suitable electrode, the voltammetric performances of different carbon electrodes were evaluated. Two widely used $sp^2$ carbon-based electrodes (GCE and SPCE) were selected and compared with BDDE. Because of its unique $sp^3$ diamond structure, BDDE manifests various superior properties for electrochemical-sensing applications, including the wide electrochemical potential window, low


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background current, high fouling resistance and biocompatibility, relatively rapid electron transfer kinetics, and long-term stability under high-potential operation.[24,25]

The electrochemical potential windows of the selected carbon electrodes were first characterized to define the suitable potential window for electroanalysis while avoiding the electrolysis of the biofluid. Because of the potentially high oxidation potential of the target drug molecules, and broadly, to enable the analysis of a larger panel of electroactive drugs, a wide potential window is desired. Here, the potential window was defined with the aid of CV and by setting the cutoff current density as 50 $\mu$A/cm$^2$.[26] As shown in Figure 2a, BDDE exhibits the widest potential window as compared to GCE and SPCE, which arises from its high overpotential for solution decomposition. Within the electrochemical potential window, a low and stable background current is desired for high signal-to-background ratio voltammetric analysis. The background current level is related to the double-layer capacitance at the electrode−electrolyte interface, which can be effectively characterized by CV measurement.[27] Figure S2 shows that the double-layer capacitance of the tested BDDE was ∼8 $\mu$F/cm$^2$, which was less than 1/3 of that of the GCE, implying that a lower background current is to be expected when performing voltammetric measurements with BDDE.[25]

Next, the stability of background current under long-term operation at a high potential was evaluated across the three carbon electrodes. Figure 2b shows that, under a constant voltage of 1.2 V, progressive baseline current drift occurred in the cases of GCE and SPCE, which can be attributed to the surface oxidation reaction of sp$^2$ carbon.[28] Conversely, no significant baseline current change was observed for the case of BDDE, which can be attributed to its microstructural stability and resistance to oxidative damage.[24] These results demonstrate the superior long-term stability of BDDE even at a high-voltage operation.

Given its desirable properties for voltammetry-based drug analysis (as characterized and discussed above), hereafter, BDDE is selected to target DP, APAP, and CAFF as three model electroactive drugs. The electrochemical response of BDDE can be tuned by modifying its surface termination to optimize its quantification capability toward the target drugs.[29] The surface of the as-deposited BDDE is typically hydrogen-terminated, whereas anodic treatment can effectively alter the surface to be oxygen-terminated.[27,30] As the surface termination of BDDE can affect the electrolyte organization at the electrode−electrolyte interface and the electron transfer kinetics, the suitability of the surface chemistry must be examined in relation to each drug. Here, the redox responses of the three target drug molecules in their physiological relevant concentration range were characterized by CV using anodic-treated and untreated BDDEs. As shown in Figure 2c, for anodic-treated BDDE (oxygen-terminated), three well-separated oxidation peaks were observed in the cyclic voltammogram, whereas for the case of the untreated electrode (hydrogen-terminated), only APAP presented a clear electro-oxidation response with a small overlapped DP oxidation peak on its left shoulder. The oxidation of CAFF on untreated BDDE is not evident in the voltammogram, because of the high current background level. These observations illustrate the utility of the anodic-treated BDDE to render simultaneous and high-sensitivity quantification of the target drug molecules.

Among the common voltammetry techniques, here, DPV was adopted for the quantification of target drugs, because it allows for mitigating the contribution of non-faradaic background, and thus achieving high signal-to-background ratio measurements. The voltammograms of DP, APAP, and CAFF (for different concentration levels, within their physiological relevant range) are shown in Figures 2d−f and S3. The calibration curves were shown in the insets, respectively, where the peak current information was extracted with the aid of the background (measurement without analyte) subtraction. These results demonstrate that oxidation peaks can be recorded in nano- to sub/low-micromolar level for all the model drugs, demonstrating the suitability of the adapted electroanalytical sensing approach for the envisioned applications, in terms of meeting the limit of detection criterion. To provide a measure of comparison, the responses of target drugs were also recorded by SPCE and GCE. As shown in Figure S4, generally, BDDE rendered measurements with a higher signal-to-background ratio, as compared to SPCE and GCE, validating BDDE's superior properties for the context at hand.

**Interference Study To Determine the Operational Window for Electroactive Drug Analysis.** To adapt the devised sensing approach for the analysis of complex and untreated biofluid samples, the interference of nontarget species must be characterized. For the context at hand, specifically, the voltammetric response of endogenous electroactive species within the biofluid can lead to the burial of the target drug's voltammetric characteristic peak. Accordingly, five highly electroactive model interferents (with widely spread oxidation potential) are selected,[31−34] which are present in human sweat at relatively high and with dynamically varying concentration levels.[35−39] The interferents include amino acids (tryptophan, TRY, tyrosine, TYR, methionine, MET, and histidine, HIS) and UA, which are listed in Figure 3a, along with their previously characterized concentration ranges in sweat.[35,39]

The contribution of each electroactive interferent to the voltammetric background current was first investigated and characterized. As can be seen in Figures 3a and S5, upon the introduction of each interferent in a test buffer (acetate buffer, pH = 5), the corresponding voltammetric background currents were significantly distorted (because of the respective interferent's oxidation reaction). To quantify the relative distortion of the background current by each of the interferents, a potential window of influence is defined, which captures the potential range where the spiked interferent causes greater than 50% change to the original background current. The observed extended potential windows of influence for the tested amino acids (spanning from the respective oxidation peak positions to relatively high potential levels, >1.2 V) can be attributed to the complex cascade mechanisms involved in the oxidation of the amino acids as well as the reactants' or products' adsorption.[33,40]

To determine the potential ranges unaffected by the interferents (namely, undistorted potential windows), which present opportunities for selective drug detection, the characterized potential windows of influence were annotated in relation to the voltammetric response of the model drug molecules. As illustrated in Figure 3b, the undistorted potential window in our context is 0−0.5 V, which can be uniquely exploited to target DP. To verify this point, the voltammograms of solutions (acetate buffer, pH = 5) containing the model drugs alone were obtained and compared with that of the respective solutions spiked with the interferents oxidized at proximal potential (of the model drug). As shown in Figure 3c,


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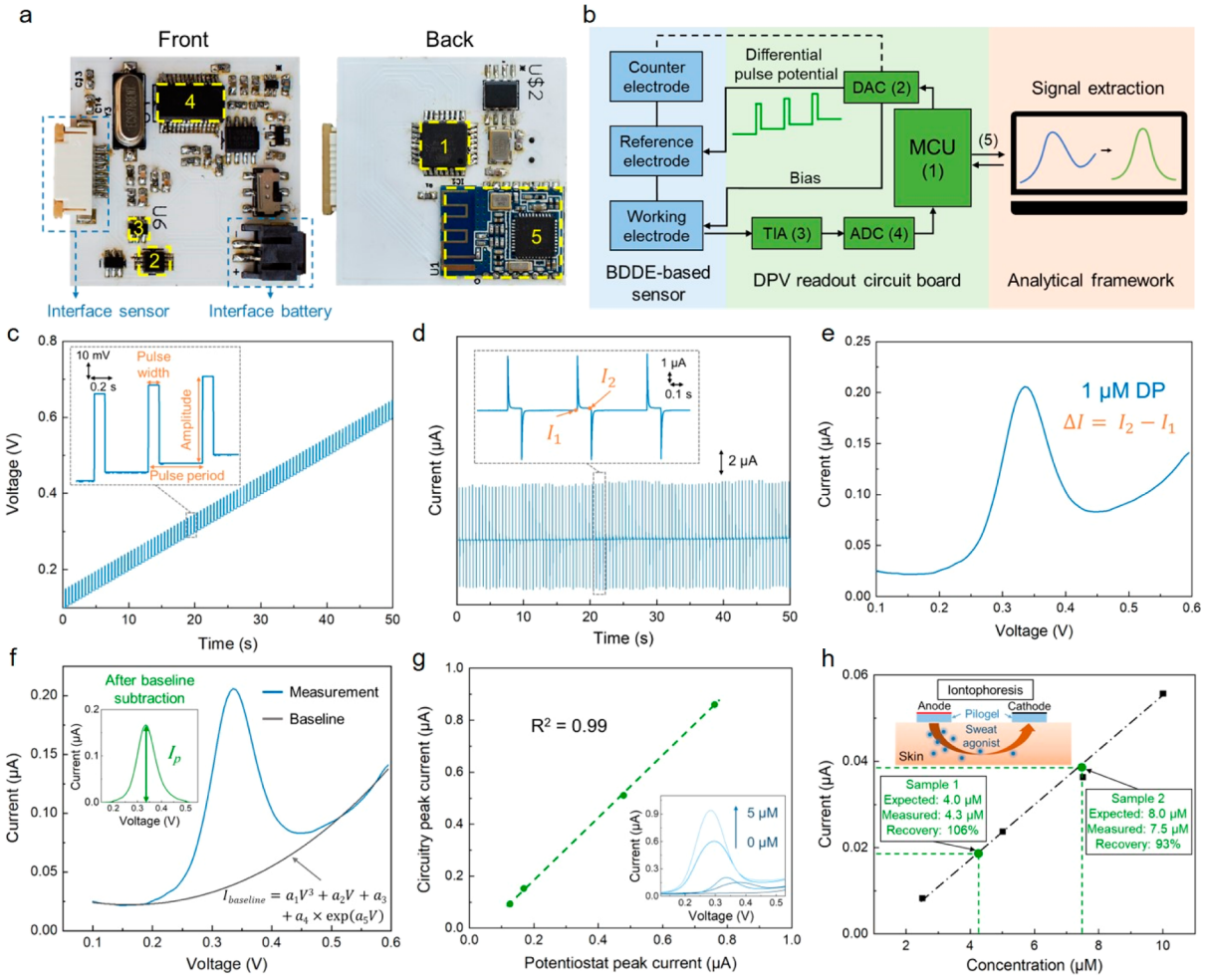



**Figure 4.** (a) Front and back views of the custom-developed DPV readout circuit board, which integrates commercially available chips, including (1) MCU, (2) DAC, (3) TIA, (4) ADC, and (5) bluetooth transceiver module. (b) System-level block diagram of the DPV readout circuit board interfacing an analytical framework. (c) Representative circuit-rendered differential pulse potential waveform (the inset shows the zoomed view, pulse period: 0.5 s, pulse width: 0.1 s, amplitude: 50 mV, step: 5 mV, potential range: 0.1−0.6 V). (d) Circuit-rendered real-time current response toward 1 $\mu$M DP (in artificial sweat, pH = 5). The inset shows the zoomed view, with an annotated pair of sampling points for differential current response calculation. (e) Differential pulse voltammogram of an artificial sweat sample (pH = 5) spiked with 1 $\mu$M DP, rendered by the devised sensing system. (f) Background current fitting of the 1 $\mu$M DP measurement (in artificial sweat, pH = 5), following a combined third-order polynomial and exponential relationship. The inset shows the baseline-subtracted voltammogram. (g) Voltammetric peak current levels of 0, 0.5, 1, 3, 5 $\mu$M DP in artificial sweat (pH = 5) measured by the developed DPV readout circuit board vs potentiostat. The inset shows the corresponding differential pulse voltammograms rendered by the board. (h) DP quantification in spiked iontophoretically induced sweat samples. The DP concentration of the test samples were estimated with the aid of the calibration curve constructed from the DP-spiked calibration samples. The inset shows the schematic of iontophoresis for sweat induction.

the fingerprint peak of DP was still visible after the addition of interferents. On the other hand, as shown in Figure 3d,e, the fingerprint peaks of APAP and CAFF were buried after the introduction of the interferents. Furthermore, voltammetric drug quantification was performed in artificial sweat (pH = 5) as a test buffer, which in addition to the characterized interferents include other amino acids, metabolites, and minerals (in their physiological relevant concentration), mimicking a complex biofluid matrix. Comparing the voltammograms of the blank artificial sweat (as background) and drug-spiked artificial sweat samples yields a distinguishable peak corresponding to the DP oxidation, whereas the oxidation peaks of the other two target drug molecules were buried in the background (Figure 3f).

Whereas these results suggest that voltammetric quantification in the distorted potential window (here, applicable to APAP and CAFF) warrants additional efforts (e.g., sensor surface modification and/or calibration via a dedicated analytical framework), they inform the feasibility of DP quantification with the readily devised and described voltammetric sensing interface. To further characterize the suitability of the approach for DP quantification, complementary experiments were performed to capture the effect of nonelectroactive interferents and pH variations on the BDDE response. The results of the selectivity study toward nonelectroactive species are shown in Figure S6, illustrating the minimal interference of those interferents. The BDDE response toward DP in different pH conditions within the physiological range of sweat (pH = 5−7) is shown in Figure S7. The observed negative shift of the oxidation peak voltage as a result of pH increase from 5 to 7 can be attributed to the involvement of proton transfer in the electrochemical reaction.[16] In all the pH conditions, the observed oxidation peak voltages were within the undistorted potential window,


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illustrating the capability of DP quantification by the devised sensing interface. The decrease of the DP peak current is related to the analyte and electrode surface interaction, informing the need for incorporating an auxiliary pH-sensing interface for sensor response calibration to account for the variation of pH in sweat. Furthermore, voltammetric DP calibration was performed in artificial sweat with the devised sensing interface. As shown in Figure 3g, the results indicate that 50 nM limit of detection can be achieved in a complex matrix, which is similar to that achieved in the acetate buffer solution containing the target only.

**Wireless Voltammetric Sensing System for Wearable Electroactive Drug Analysis.** The findings from the studies performed within the presented framework (including establishing the appropriate sensing electrode and electroanalytical method as well as defining the operational window for undistorted voltammetric analysis) can be leveraged to design a voltammetric BDDE-based sensing system to render sample-to-answer drug readouts. To this end, circuit-level DPV-based signal excitation/acquisition as well as wireless transmission capabilities need to be developed and integrated in a small footprint. In this context, the excitation signal is a differential pulse potential waveform (potential pulses superimposed upon a staircase waveform), which is applied across the working and reference electrodes. The sensor current output can be acquired with the aid of a transimpedance stage, which converts the current signal into voltage.

Accordingly, a bluetooth-enabled DPV circuit board is custom-developed (Figure 4a), which implements the required system-level operations (Figure 4b) seamlessly, following the circuit specifications informed by the aforementioned characterizations with the potentiostat. At its core, the circuit board utilizes an MCU to program the required excitation potential waveform (with the aid of a DAC) and to retrieve the TIA-processed readout signal (with the aid of a peripheral ADC). Figure 4c,d illustrates a representative circuit-rendered differential pulse potential waveform and a real-time continuous current readout, respectively. The difference of the averaged current before upward and before downward voltage change (annotated in Figure 4d,e) was used to construct the DPV voltammogram (Figure 4e) from the recorded current response.

To accurately extract the peak height information from the voltammogram, the DPV readouts must be augmented with an analytical framework for background current estimation and correction. In that regard, here a combined third-order polynomial and exponential relationship was utilized to estimate the background current, similar to previous efforts.[41] The versatility of this function has been proven to be useful in providing an accurate estimation of the current background, which is particularly advantageous over linear fitting when quantifying low-concentration target analytes where the analyte oxidation peak is likely to be distorted by a rising/descending background. Accordingly, the residual measured current outside the peak window was first fitted to the proposed relationship to generate the estimated background current.[42] Then, as illustrated in Figure 4f, the peak height information was extracted by subtracting the derived background from the current measurements. The circuitry-based readings were subsequently validated by comparing the background-corrected measurements with that obtained by the lab instrument ($R^2$ = 0.99, Figure 4g).

To demonstrate the utility of the devised approach to render sample-to-answer drug readouts in epidermally retrievable biofluids, the DP concentration levels in spiked sweat samples were quantified by the developed sensing system and analytical framework. Accordingly, standard iontophoresis protocol[43] was followed to obtain a sweat sample from a healthy human subject without DP administration history (hence expecting zero amount of DP in the sweat sample). The obtained sweat sample was aliquoted and spiked with DP to construct four calibration samples and two test samples (4.0 and 8.0 $\mu$M). The BDDE-based voltammetric sensing system and analytical framework was then adapted to analyze the four calibration samples (Figure S8a) in order to construct the response curve (Figure 4h). It is worth noting that the difference of the sensor response in the actual versus artificial sweat samples can be attributed to the difference in the composition of the two matrices. The response curve constructed by the spiked sweat samples was used to infer the DP concentration levels of the subsequently introduced two test samples (Figure S8b,c). As shown in Figure 4h, the estimated DP concentration levels for the respective test samples were 4.3 and 7.5 $\mu$M (representing <10% error). These results further validate the capability of the developed BDDE-based voltammetric sensing system and analytical framework for accurate electroactive drug quantification in complex biofluids such as sweat.

## CONCLUSIONS AND OUTLOOK

In conclusion, the design space and operational settings informed by the presented framework provided the basis for the development of a coupled sensing system and analytical framework for rendering electroactive drug readouts in a sample-to-answer manner. In that regard, the evaluation of carbon-based electrode candidates informed the suitability of BDDE for the envisioned application, which, followed by anodic treatment, was utilized to demonstrate nano- to sub/low-micromolar level quantification of three model drugs (using DPV-based analysis). Furthermore, the characterization of the influence of the electroactive species present in epidermally retrievable biofluids (including amino acids and UA) on the sensor response allowed for the definition/identification of undistorted potential window(s) (<0.5 V in our context) for the selective quantification of target electroactive molecules with oxidation peaks residing within the identified window(s). Accordingly, this work establishes, for the first time (to the best of our knowledge), the utility of BDDE toward targeting electroactive species in epidermally retrievable biofluids by performing a comprehensive electrode characterization and interference study relevant to sweat. The findings from such study were particularly leveraged to render DP concentration readouts in iontophoretically induced sweat samples using a coupled sensing system and analytical framework.

Future work will focus on mitigating the influence of endogenous electroactive interferents, by devising surface modification techniques and a dedicated analytical framework in order to enable the application of the demonstrated methodologies for the quantification of a wider panel of analytes (such as those with oxidation peaks residing in distorted potential windows). Furthermore, to adapt the presented methodologies for wearable sweat-based analysis, an iontophoresis interface should be integrated[44] to allow sweat induction on-demand in sedentary subjects. Additionally, the inclusion of auxiliary sensing interfaces for in situ sensor


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calibration (e.g., pH) and characterization of sweat secretion profile (e.g., sweat rate, volume loss, etc.) may be helpful in normalizing the readings against inter/intrasubject physiological variations and gland activity variability. Moreover, surface regeneration methods may need to be devised[45,46] to mitigate undesired sensor surface fouling and contamination effects toward long-term usage (minimizing the user intervention in terms of the frequency of the sensor replacement). Seamless integration of these functionalities within a BDDE-based voltammetric sensing system—packaged in a wearable casing—is necessary for the realization of fully autonomous operation and adoption for translational applications. In that regard, in parallel to such engineering efforts and to inform clinical utility, large-scale clinical investigations need to be performed to systematically characterize the correlation of target electroactive drug molecules (e.g., DP) in sweat versus blood in relation to biological confounders (e.g., inter/intraindividual variations), toward constructing the target's equivalent sweat-based PK profile.

Overall, the presented design framework and wireless voltammetric sensing system can serve as a basis for the future wearable sensor development efforts for applications such as pharmaceutical/drug abuse monitoring for personalized therapy management.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssensors.9b02233.

> Microfluidic module; comparison of the double-layer capacitance of BDDE and GCE; DP calibration in a relatively low concentration range; comparison of the response of SPCE, GCE, and BDDE toward model drugs; voltammetric response of electroactive interferents within their physiological relevant concentration ranges in sweat; interference study of nonelectroactive species; pH interference of DP response; and differential pulse voltammograms of spiked sweat samples (PDF)

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S.L. and B.W. contributed equally. S.L., B.W. and S.E. conceived the idea and designed the experiments; S.L. led the experiments (with assistance from B.W., W.Y., K.C., C.H., X.C., Y.Z., Y.G., Z.W., H.L., H.H., J.T., and S.E.); W.Y. led PCB and system design; S.L., B.W., W.Y., K.C., C.H., X.C., Y.Z., Y.G., Z.W., H.L., H.H., J.T., and S.E. contributed to data analysis and interpretation; S.L., B.W., W.Y., and S.E. drafted the paper and all authors provided feedback. S.E. supervised the study.

### Notes

The authors declare no competing financial interest.

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## ■ REFERENCES


(1) Heikenfeld, J.; Jajack, A.; Rogers, J.; Gutruf, P.; Tian, L.; Pan, T.; Li, R.; Khine, M.; Kim, J.; Wang, J. Wearable Sensors: Modalities, Challenges, and Prospects. *Lab Chip* **2018**, *18*, 217−248.

(2) Bariya, M.; Nyein, H. Y. Y.; Javey, A. Wearable Sweat Sensors. *Nat. Electron.* **2018**, *1*, 160−171.

(3) Lin, H.; Hojaiji, H.; Lin, S.; Yeung, C.; Zhao, Y.; Wang, B.; Malige, M.; Wang, Y.; King, K.; Yu, W. Wearable Electrofluidic Actuation System. *Lab Chip* **2019**, *19*, 2966−2972.

(4) Lin, S.; Wang, B.; Zhao, Y.; Shih, R.; Cheng, X.; Yu, W.; Hojaiji, H.; Lin, H.; Hoffman, C.; Ly, D. Natural Perspiration Sampling and in Situ Electrochemical Analysis with Hydrogel Micropatches for User-Identifiable and Wireless Chemo/Biosensing. *ACS Sens.* **2019**, DOI: 10.1021/acssensors.9b01727.

(5) Gao, W.; Emaminejad, S.; Nyein, H. Y. Y.; Challa, S.; Chen, K.; Peck, A.; Fahad, H. M.; Ota, H.; Shiraki, H.; Kiriya, D.; et al. Fully Integrated Wearable Sensor Arrays for Multiplexed in Situ Perspiration Analysis. *Nature* **2016**, *529*, 509−514.

(6) Kim, J.; Campbell, A. S.; de Ávila, B. E.-F.; Wang, J. Wearable Biosensors for Healthcare Monitoring. *Nat. Biotechnol.* **2019**, *37*, 389−406.

(7) Koh, A.; Kang, D.; Xue, Y.; Lee, S.; Pielak, R. M.; Kim, J.; Hwang, T.; Min, S.; Banks, A.; Bastien, P. A Soft, Wearable Microfluidic Device for the Capture, Storage, and Colorimetric Sensing of Sweat. *Sci. Transl. Med.* **2016**, *8*, 366ra165.

(8) Cheng, X.; Wang, B.; Zhao, Y.; Hojaiji, H.; Lin, S.; Shih, R.; Lin, H.; Tamayosa, S.; Ham, B.; Stout, P.; Salahi, K.; Wang, Z.; Zhao, C.; Tan, J.; Emaminejad, S. A Mediator-Free Electroenzymatic Sensing Methodology to Mitigate Ionic and Electroactive Interferents' Effects for Reliable Wearable Metabolite and Nutrient Monitoring. *Adv. Funct. Mater.* **2019**, 1908507.

(9) Heikenfeld, J.; Jajack, A.; Feldman, B.; Granger, S. W.; Gaitonde, S.; Begtrup, G.; Katchman, B. A. Accessing Analytes in Biofluids for Peripheral Biochemical Monitoring. *Nat. Biotechnol.* **2019**, *37*, 407−419.

(10) Sonner, Z.; Wilder, E.; Heikenfeld, J.; Kasting, G.; Beyette, F.; Swaile, D.; Sherman, F.; Joyce, J.; Hagen, J.; Kelley-Loughnane, N. The Microfluidics of the Eccrine Sweat Gland, Including Biomarker Partitioning, Transport, and Biosensing Implications. *Biomicrofluidics* **2015**, *9*, 031301.

(11) Kovacs, E. M. R.; Stegen, J. H. C. H.; Brouns, F. Effect of Caffeinated Drinks on Substrate Metabolism, Caffeine Excretion, and Performance. *J. Appl. Physiol.* **1998**, *85*, 709−715.

(12) Bard, A. J.; Faulkner, L. R.; Leddy, J.; Zoski, C. G. *Electrochemical Methods: Fundamentals and Applications*; wiley: New York, 1980; Vol. 2.

(13) Clarke, W.; Dasgupta, A. *Clinical Challenges in Therapeutic Drug Monitoring: Special Populations, Physiological Conditions and Pharmacogenomics*; Elsevier, 2016.

(14) Švorc, L.; Tomčík, P.; Svítková, J.; Rievaj, M.; Bustin, D. Voltammetric Determination of Caffeine in Beverage Samples on Bare Boron-Doped Diamond Electrode. *Food Chem.* **2012**, *135*, 1198−1204.
(15) Lourenção, B. C.; Medeiros, R. A.; Rocha-Filho, R. C.; Mazo, L. H.; Fatibello-Filho, O. Simultaneous Voltammetric Determination of Paracetamol and Caffeine in Pharmaceutical Formulations Using a Boron-Doped Diamond Electrode. *Talanta* **2009**, *78*, 748−752.
(16) Sarakhman, O.; Pysarevska, S.; Dubenska, L.; Stanković, D. M.; Otřísal, P.; Planková, A.; Kianičková, K.; Švorc, Ľ. Voltammetric Protocol for Reliable Determination of a Platelet Aggregation Inhibitor Dipyridamole on a Bare Miniaturized Boron-Doped Diamond Electrochemical Sensor. *J. Electrochem. Soc.* **2019**, *166*, B219−B226.
(17) Schaper, W. Dipyridamole, an Underestimated Vascular Protective Drug. *Cardiovasc. Drugs Ther.* **2005**, *19*, 357−363.
(18) Rao, G. H. R. Need for a Point-of-Care Assay for Monitoring Antiplatelet and Antithrombotic Therapies. *Stroke* **2009**, *40*, 2271−2272.
(19) Flint, R. B.; Mian, P.; van der Nagel, B.; Slijkhuis, N.; Koch, B. C. P. Quantification of Acetaminophen and Its Metabolites in Plasma Using UPLC-MS: Doors Open to Therapeutic Drug Monitoring in Special Patient Populations. *Ther. Drug Monit.* **2017**, *39*, 164−171.
(20) de Wildt, S. N.; Kerkvliet, K. T. M.; Wezenberg, M. G. A.; Ottink, S.; Hop, W. C. J.; Vulto, A. G.; van den Anker, J. N. Use of Saliva in Therapeutic Drug Monitoring of Caffeine in Preterm Infants. *Ther. Drug Monit.* **2001**, *23*, 250−254.
(21) Gal, P. Caffeine Therapeutic Drug Monitoring Is Necessary and Cost-Effective. *J. Pediatr. Pharmacol. Ther.* **2007**, *12*, 212−215.
(22) Tai, L.-C.; Gao, W.; Chao, M.; Bariya, M.; Ngo, Q. P.; Shahpar, Z.; Nyein, H. Y. Y.; Park, H.; Sun, J.; Jung, Y.; et al. Methylxanthine Drug Monitoring with Wearable Sweat Sensors. *Adv. Mater.* **2018**, *30*, 1707442.
(23) Gupta, V. K.; Jain, R.; Radhapyari, K.; Jadon, N.; Agarwal, S. Voltammetric techniques for the assay of pharmaceuticals-A review. *Anal. Biochem.* **2011**, *408*, 179−196.
(24) Yang, N.; Yu, S.; Macpherson, J. V.; Einaga, Y.; Zhao, H.; Zhao, G.; Swain, G. M.; Jiang, X. Conductive Diamond: Synthesis, Properties, and Electrochemical Applications. *Chem. Soc. Rev.* **2019**, *48*, 157−204.
(25) Macpherson, J. V. A Practical Guide to Using Boron Doped Diamond in Electrochemical Research. *Phys. Chem. Chem. Phys.* **2015**, *17*, 2935−2949.
(26) Benck, J. D.; Pinaud, B. A.; Gorlin, Y.; Jaramillo, T. F. Substrate Selection for Fundamental Studies of Electrocatalysts and Photoelectrodes: Inert Potential Windows in Acidic, Neutral, and Basic Electrolyte. *PLoS One* **2014**, *9*, No. e107942.
(27) Hutton, L. A.; Iacobini, J. G.; Bitziou, E.; Channon, R. B.; Newton, M. E.; Macpherson, J. V. Examination of the Factors Affecting the Electrochemical Performance of Oxygen-Terminated Polycrystalline Boron-Doped Diamond Electrodes. *Anal. Chem.* **2013**, *85*, 7230−7240.
(28) Xu, J.; Swain, G. M. Oxidation of Azide Anion at Boron-Doped Diamond Thin-Film Electrodes. *Anal. Chem.* **1998**, *70*, 1502−1510.
(29) Hupert, M.; Muck, A.; Wang, J.; Stotter, J.; Cvackova, Z.; Haymond, S.; Show, Y.; Swain, G. M. Conductive Diamond Thin-Films in Electrochemistry. *Diamond Relat. Mater.* **2003**, *12*, 1940−1949.
(30) Ivandini, T. A.; Rao, T. N.; Fujishima, A.; Einaga, Y. Electrochemical Oxidation of Oxalic Acid at Highly Boron-Doped Diamond Electrodes. *Anal. Chem.* **2006**, *78*, 3467−3471.
(31) Wang, Q.; Vasilescu, A.; Subramanian, P.; Vezeanu, A.; Andrei, V.; Coffinier, Y.; Li, M.; Boukherroub, R.; Szunerits, S. Simultaneous Electrochemical Detection of Tryptophan and Tyrosine Using Boron-Doped Diamond and Diamond Nanowire Electrodes. *Electrochem. Commun.* **2013**, *35*, 84−87.
(32) Chen, L.-C.; Chang, C.-C.; Chang, H.-C. Electrochemical Oxidation of Histidine at an Anodic Oxidized Boron-Doped Diamond Electrode in Neutral Solution. *Electrochim. Acta* **2008**, *53*, 2883−2889.
(33) Enache, T. A.; Oliveira-Brett, A. M. Boron Doped Diamond and Glassy Carbon Electrodes Comparative Study of the Oxidation Behaviour of Cysteine and Methionine. *Bioelectrochemistry* **2011**, *81*, 46−52.
(34) Popa, E.; Kubota, Y.; Tryk, D. A.; Fujishima, A. Selective Voltammetric and Amperometric Detection of Uric Acid with Oxidized Diamond Film Electrodes. *Anal. Chem.* **2000**, *72*, 1724−1727.
(35) Delgado-Povedano, M. M.; Calderón-Santiago, M.; Priego-Capote, F.; de Castro, M. D. L. Study of sample preparation for quantitative analysis of amino acids in human sweat by liquid chromatography-tandem mass spectrometry. *Talanta* **2016**, *146*, 310−317.
(36) Liappis, N.; Hungerland, H. Quantitative Study of Free Amino Acids in Human Eccrine Sweat during Normal Conditions and Exercise. *Am. J. Clin. Nutr.* **1972**, *25*, 661−663.
(37) Liappis, N.; Kelderbacher, S.-D.; Kesseler, K.; Bantzer, P. Quantitative Study of Free Amino Acids in Human Eccrine Sweat Excreted from the Forearms of Healthy Trained and Untrained Men during Exercise. *Eur. J. Appl. Physiol. Occup. Physiol.* **1979**, *42*, 227−234.
(38) Mark, H.; Harding, C. R. Amino Acid Composition, Including Key Derivatives of Eccrine Sweat: Potential Biomarkers of Certain Atopic Skin Conditions. *Int. J. Cosmet. Sci.* **2013**, *35*, 163−168.
(39) Huang, C.-T.; Chen, M.-L.; Huang, L.-L.; Mao, I.-F. Uric Acid and Urea in Human Sweat. *Chin. J. Physiol.* **2002**, *45*, 109−115.
(40) Zhao, G.; Qi, Y.; Tian, Y. Simultaneous and Direct Determination of Tryptophan and Tyrosine at Boron-Doped Diamond Electrode. *Electroanalysis* **2006**, *18*, 830−834.
(41) Omanović, D.; Garnier, C.; Louis, Y.; Lenoble, V.; Mounier, S.; Pižeta, I. Significance of Data Treatment and Experimental Setup on the Determination of Copper Complexing Parameters by Anodic Stripping Voltammetry. *Anal. Chim. Acta* **2010**, *664*, 136−143.
(42) Jakubowska, M. Signal Processing in Electrochemistry. *Electroanalysis* **2011**, *23*, 553−572.
(43) LeGrys, V. A.; Burritt, M. F.; Gibson, L. E.; Hammond, K. B., Kraft, K.; Rosenstein, B. J. *Sweat testing: sample collection and quantitative analysis. Approved Guideline*; National Committee for Clinical Laboratory Standards, 2000; Vol. *20*.
(44) Emaminejad, S.; Gao, W.; Wu, E.; Davies, Z. A.; Nyein, H. Y. Y.; Challa, S.; Ryan, S. P.; Fahad, H. M.; Chen, K.; Shahpar, Z. Sweat Extraction and Analysis Applied to Cystic Fibrosis and Glucose Monitoring Using a Fully Integrated Wearable Platform. *Proc. Natl. Acad. Sci. U.S.A.* **2017**, *114*, 4625−4630.
(45) Kiran, R.; Scorsone, E.; Mailley, P.; Bergonzo, P. Quasi-Real Time Quantification of Uric Acid in Urine Using Boron Doped Diamond Microelectrode with in Situ Cleaning. *Anal. Chem.* **2012**, *84*, 10207−10213.
(46) Manica, D. P.; Mitsumori, Y.; Ewing, A. G. Characterization of Electrode Fouling and Surface Regeneration for a Platinum Electrode on an Electrophoresis Microchip. *Anal. Chem.* **2003**, *75*, 4572−4577.


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